

36-226 Notes Set 3: Properties of Maximum Likelihood Estimators

Combined typed notes from the handout and handwritten annotations

Optional associated reading: Wackerly 9.7–9.8

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1 Big Picture

These notes discuss useful properties of maximum likelihood estimators (MLEs):

- **Invariance:** the MLE of a function of a parameter is the function of the MLE.
- **Asymptotic unbiasedness:** MLEs may be biased for finite n , but often become unbiased in the limit.
- **Asymptotic normality:** for large samples, MLEs are approximately Normal.
- **Asymptotic minimum variance:** MLEs asymptotically attain the Cramer–Rao lower bound under regularity conditions.
- **Exponential families:** many common distributions have a form that makes sufficiency easy to identify.

Key point

MLE problems usually reduce to one of two patterns:

1. **Interior MLE:** differentiate the log-likelihood, set the score equal to 0, solve.
2. **Boundary/support MLE:** if the support depends on the parameter, the MLE often occurs at the smallest or largest parameter value allowed by the data.

2 Invariance Property of the MLE

Suppose we have derived the MLE $\hat{\theta}$ for θ , and we want the MLE of a function of the parameter, $t(\theta)$.

Key point

Invariance theorem. If $\hat{\theta}$ is the MLE of θ , then the MLE of $t(\theta)$ is

$$\widehat{t(\theta)}_{\text{MLE}} = t(\hat{\theta}_{\text{MLE}}).$$

In words: **plug the MLE into the function.**

This is very useful because once you find $\hat{\theta}_{\text{MLE}}$, you automatically get the MLE of quantities like θ^2 , e^θ , $1/\theta$, $\text{Var}(Y)$, etc.

2.1 Proof for one-to-one transformations

Recall that t is one-to-one if it never takes the same value twice:

$$\theta_1 \neq \theta_2 \implies t(\theta_1) \neq t(\theta_2),$$

or equivalently,

$$t(\theta_1) = t(\theta_2) \implies \theta_1 = \theta_2.$$

Let

$$\eta = t(\theta).$$

If t is one-to-one, then it has an inverse:

$$\theta = t^{-1}(\eta).$$

We want the $\hat{\eta}$ that maximizes the likelihood written as a function of η :

$$L(y \mid \eta) = L(y \mid t^{-1}(\eta)).$$

We already know $\hat{\theta}$ maximizes $L(y \mid \theta)$, so the maximum likelihood value is

$$L(y \mid \hat{\theta}).$$

To achieve this same maximum in the η parameterization, choose $\hat{\eta}$ so that

$$t^{-1}(\hat{\eta}) = \hat{\theta}.$$

Thus,

$$\hat{\eta} = t(\hat{\theta}).$$

Therefore, the MLE of $\eta = t(\theta)$ is $t(\hat{\theta})$.

2.2 Example: Exponential variance

Let

$$Y_1, \dots, Y_n \stackrel{\text{i.i.d.}}{\sim} \text{Exp}(\beta), \quad f_Y(y) = \frac{1}{\beta} e^{-y/\beta}, \quad y, \beta > 0.$$

For this parameterization,

$$\text{E}[Y] = \beta, \quad \text{Var}(Y) = \beta^2.$$

From earlier notes,

$$\hat{\beta}_{\text{MLE}} = \bar{Y}.$$

By invariance,

$$\widehat{\text{Var}(Y)}_{\text{MLE}} = \widehat{\beta^2}_{\text{MLE}} = (\hat{\beta}_{\text{MLE}})^2 = \bar{Y}^2.$$

2.3 Example: Uniform distribution with parameter-dependent support

Let

$$Y_1, \dots, Y_n \stackrel{\text{i.i.d.}}{\sim} \text{Unif}[0, 2\theta + 1], \quad f_Y(y) = \frac{1}{2\theta + 1}, \quad 0 \leq y \leq 2\theta + 1.$$

We know

$$\text{E}[Y] = \frac{2\theta + 1 + 0}{2} = \theta + \frac{1}{2}, \quad \text{Var}(Y) = \frac{(2\theta + 1)^2}{12}.$$

The likelihood is

$$L(y \mid \theta) = \prod_{i=1}^n \frac{1}{2\theta + 1} = \frac{1}{(2\theta + 1)^n},$$

with the constraint

$$0 \leq y_i \leq 2\theta + 1 \quad \text{for all } i.$$

The likelihood decreases as θ increases, so we want θ as small as possible while still satisfying the data constraint. The constraint says

$$\max_i y_i \leq 2\theta + 1.$$

Let

$$Y_{(n)} = \max(Y_1, \dots, Y_n).$$

Then

$$Y_{(n)} \leq 2\theta + 1 \iff \frac{Y_{(n)} - 1}{2} \leq \theta.$$

Therefore the smallest allowed value is

$$\hat{\theta}_{\text{MLE}} = \frac{Y_{(n)} - 1}{2}.$$

By invariance, the MLE of $\text{Var}(Y)$ is

$$\widehat{\text{Var}(Y)}_{\text{MLE}} = \frac{(2\hat{\theta}_{\text{MLE}} + 1)^2}{12} = \frac{\left(2 \cdot \frac{Y_{(n)} - 1}{2} + 1\right)^2}{12} = \frac{Y_{(n)}^2}{12}.$$

Careful

When the support depends on the parameter, do **not** blindly differentiate. First write the likelihood with its constraint. Here the likelihood is largest at a boundary because making θ smaller makes $(2\theta + 1)^{-n}$ larger.

3 Asymptotic Unbiasedness

Key point

Let $\hat{\theta}_n$ denote the MLE of θ based on i.i.d. samples $Y_1, \dots, Y_n$. Under regularity conditions, the MLE is **asymptotically unbiased**:

$$\lim_{n \rightarrow \infty} \mathbb{E}[\hat{\theta}_n] = \theta.$$

This does **not** mean the MLE is unbiased for finite n .

3.1 Example: Normal variance MLE

Let

$$Y_1, \dots, Y_n \stackrel{\text{i.i.d.}}{\sim} N(\mu, \sigma^2).$$

The MLE of σ^2 is

$$\hat{\sigma}_{\text{MLE}}^2 = \frac{1}{n} \sum_{i=1}^n (Y_i - \bar{Y})^2.$$

The usual unbiased sample variance is

$$S^2 = \frac{1}{n-1} \sum_{i=1}^n (Y_i - \bar{Y})^2.$$

Therefore

$$\hat{\sigma}_{\text{MLE}}^2 = \frac{n-1}{n} S^2.$$

Since S^2 is unbiased,

$$\mathbb{E}[S^2] = \sigma^2.$$

Thus

$$\mathbb{E}[\hat{\sigma}_{\text{MLE}}^2] = \mathbb{E}\left[\frac{n-1}{n} S^2\right] = \frac{n-1}{n} \sigma^2.$$

So $\hat{\sigma}_{\text{MLE}}^2$ is biased downward for finite n . But

$$\lim_{n \rightarrow \infty} \text{E}[\hat{\sigma}_{\text{MLE}}^2] = \lim_{n \rightarrow \infty} \frac{n-1}{n} \sigma^2 = \sigma^2.$$

So the MLE is asymptotically unbiased.

Careful

The MLE variance estimator uses denominator n , not $n-1$. The unbiased sample variance uses denominator $n-1$. Therefore

$$\hat{\sigma}_{\text{MLE}}^2 = \frac{n-1}{n} S^2,$$

not $\frac{n}{n-1} S^2$.

4 Asymptotic Normality

Let $\hat{\theta}_n$ be the MLE based on i.i.d. samples $Y_1, \dots, Y_n$ with density or pmf $f_Y(y | \theta)$.

The goal is to show that, for large n ,

$$\sqrt{n}(\hat{\theta} - \theta) \xrightarrow{d} N(0, \sigma_{\text{MLE}}^2),$$

and compute the asymptotic variance. The two central tools are the **score function** and **Fisher information**.

4.1 Log-likelihood and score function

The log-likelihood is

$$\ell(y | \theta) = \log L(y | \theta).$$

Key point

The **score function** is the derivative of the log-likelihood:

$$\ell'(y | \theta) = \frac{d}{d\theta} \ell(y | \theta).$$

For an interior MLE, setting the score equal to zero and solving for θ gives the MLE:

$$\ell'(y | \theta) = 0.$$

For one observation Y , define

$$\ell_1(Y | \theta) = \log f_Y(Y | \theta).$$

Then for the full sample,

$$\ell_n(y | \theta) = \sum_{i=1}^n \ell_1(y_i | \theta) = \sum_{i=1}^n \log f_Y(y_i | \theta).$$

4.2 Fisher information

For one observation, the Fisher information is

$$I(\theta) = \text{E} \left[(\ell'_1(Y | \theta))^2 \right].$$

For n i.i.d. observations, the Fisher information is

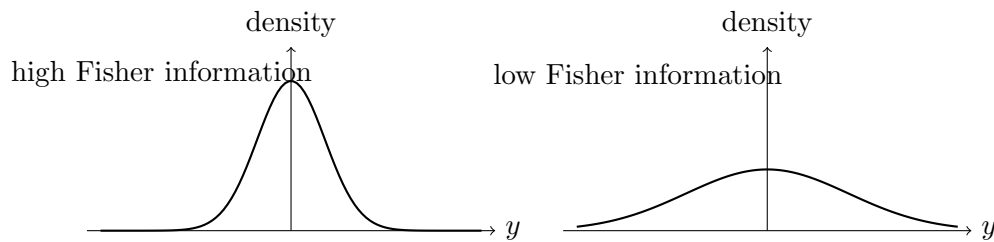
$$I_n(\theta) = nI(\theta).$$

Equivalently, using the full-sample log-likelihood,

$$I_n(\theta) = \mathbb{E} \left[(\ell'_n(Y | \theta))^2 \right].$$

Fisher information is useful because it is:

- a measure of information contained in a dataset;
- used to approximate the distribution of an MLE for large samples;
- used to find lower bounds on estimator variances.



A more peaked distribution around the parameter usually gives more information about that parameter. A flatter distribution gives less information.

4.3 Fisher information equals variance of the score

Under regularity conditions,

$$\mathbb{E}[\ell'_n(Y | \theta)] = 0.$$

Thus,

$$\text{Var}(\ell'_n(Y | \theta)) = \mathbb{E}[(\ell'_n(Y | \theta))^2] - (\mathbb{E}[\ell'_n(Y | \theta)])^2 = \mathbb{E}[(\ell'_n(Y | \theta))^2] = I_n(\theta).$$

Why is the score mean zero? For one observation,

$$\mathbb{E}[\ell'_1(Y | \theta)] = \int \frac{\partial}{\partial \theta} \log f(y | \theta) f(y | \theta) dy.$$

Since

$$\frac{\partial}{\partial \theta} \log f(y | \theta) = \frac{\frac{\partial}{\partial \theta} f(y | \theta)}{f(y | \theta)},$$

we get

$$\mathbb{E}[\ell'_1(Y | \theta)] = \int \frac{f'_\theta(y | \theta)}{f(y | \theta)} f(y | \theta) dy = \int f'_\theta(y | \theta) dy.$$

Assuming differentiation and integration can be exchanged,

$$\int f'_\theta(y | \theta) dy = \frac{\partial}{\partial \theta} \int f(y | \theta) dy = \frac{\partial}{\partial \theta} 1 = 0.$$

For n i.i.d. observations, the full score is the sum of the one-observation scores, so its expectation is still 0.

4.4 Three formulas for Fisher information

Under regularity conditions,

$$I_n(\theta) = \mathbb{E} [(\ell'_n(Y | \theta))^2]$$

$$I_n(\theta) = \text{Var}(\ell'_n(Y | \theta))$$

$$I_n(\theta) = -\mathbb{E}[\ell''_n(Y | \theta)] = -n \mathbb{E} \left[\frac{d^2}{d\theta^2} \log f_Y(Y_i | \theta) \right]$$

The third formula is often the fastest.

Method

To find Fisher information quickly:

1. Write the one-observation log-density $\log f_Y(y | \theta)$.
2. Differentiate twice with respect to θ .
3. Take negative expectation.
4. Multiply by n for the full sample.

4.5 Example: Normal mean with known variance

Let

$$Y_1, \dots, Y_n \stackrel{\text{i.i.d.}}{\sim} N(\mu, \sigma^2),$$

where σ^2 is known. The density is

$$f_Y(y | \mu) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2\sigma^2}(y-\mu)^2}.$$

The one-observation log-density is

$$\log f(y | \mu) = -\frac{1}{2} \log(2\pi\sigma^2) - \frac{1}{2\sigma^2}(y - \mu)^2.$$

Differentiate once:

$$\frac{d}{d\mu} \log f(y | \mu) = \frac{1}{\sigma^2}(y - \mu).$$

Differentiate twice:

$$\frac{d^2}{d\mu^2} \log f(y | \mu) = -\frac{1}{\sigma^2}.$$

Therefore

$$I_n(\mu) = -n \mathbb{E} \left[-\frac{1}{\sigma^2} \right] = \frac{n}{\sigma^2}.$$

4.6 Asymptotic normality theorem

Key point

Asymptotic Normality of the MLE. Let $\hat{\theta}$ be the MLE for θ based on i.i.d. samples $Y_1, \dots, Y_n$. Under regularity conditions,

$$\sqrt{n}(\hat{\theta} - \theta) \xrightarrow{d} N\left(0, \frac{1}{I(\theta)}\right).$$

Equivalently, for large n ,

$$\hat{\theta} \approx N\left(\theta, \frac{1}{nI(\theta)}\right) = N\left(\theta, \frac{1}{I_n(\theta)}\right).$$

4.7 Example: asymptotic distribution of $\hat{\mu}_{\text{MLE}}$

For

$$Y_1, \dots, Y_n \stackrel{\text{i.i.d.}}{\sim} N(\mu, \sigma^2),$$

with σ^2 known, we showed

$$I_n(\mu) = \frac{n}{\sigma^2}.$$

Therefore, for large n ,

$$\hat{\mu}_{\text{MLE}} \approx N\left(\mu, \frac{1}{I_n(\mu)}\right) = N\left(\mu, \frac{\sigma^2}{n}\right).$$

Here this is not only asymptotic: since

$$\hat{\mu}_{\text{MLE}} = \bar{Y},$$

and the sample mean of Normal variables is exactly Normal,

$$\bar{Y} \sim N\left(\mu, \frac{\sigma^2}{n}\right).$$

4.8 Example: Pareto distribution

Suppose

$$Y_1, \dots, Y_n$$

are i.i.d. from the Pareto distribution

$$f_Y(y | a, \theta) = \theta a^\theta y^{-(\theta+1)}, \quad y \geq a, \quad \theta > 1,$$

where a is known. Find the approximate distribution of the MLE for large n .

The large-sample distribution is

$$\hat{\theta}_{\text{MLE}} \approx N\left(\theta, \frac{1}{nI(\theta)}\right).$$

We compute $I(\theta)$ from the one-observation log-density:

$$\log f_Y(y | a, \theta) = \log \theta + \theta \log a - (\theta + 1) \log y.$$

Differentiate once:

$$\frac{d}{d\theta} \log f_Y(y | a, \theta) = \frac{1}{\theta} + \log a - \log y.$$

Differentiate twice:

$$\frac{d^2}{d\theta^2} \log f_Y(y | a, \theta) = -\frac{1}{\theta^2}.$$

Thus

$$I(\theta) = -\mathbb{E} \left[-\frac{1}{\theta^2} \right] = \frac{1}{\theta^2}.$$

Therefore

$$\hat{\theta}_{\text{MLE}} \approx \boxed{N \left(\theta, \frac{\theta^2}{n} \right)}.$$

5 Asymptotic Minimum Variance and the CRLB

Consider any unbiased estimator $\hat{\theta}$ of θ . Among unbiased estimators, smaller variance is better. The Cramer–Rao lower bound gives a lower bound on how small the variance can be.

Key point

Cramer–Rao Lower Bound (CRLB). Consider i.i.d. samples $Y_1, \dots, Y_n$ from $f_Y(y | \theta)$, where the range/support of Y_i does **not** depend on θ . If $\hat{\theta}$ is unbiased for θ , then

$$\text{Var}(\hat{\theta}) \geq \frac{1}{I_n(\theta)} = \frac{1}{-n \mathbb{E} \left[\frac{d^2}{d\theta^2} \log f_Y(Y_i | \theta) \right]}.$$

The CRLB is useful because:

1. If an unbiased estimator achieves the CRLB, it is an MVUE.
2. It gives a benchmark for comparing unbiased estimators.
3. It can rule out impossible unbiased estimators with variance below the bound.

The same $1/I_n(\theta)$ term appears in the asymptotic variance of the MLE. Thus, under regularity conditions, the MLE asymptotically attains the CRLB.

Careful

The CRLB has conditions. In particular, the common version here assumes the support does not depend on θ and the estimator is unbiased. It does not automatically apply to every estimator or every model.

5.1 Example: Exponential CRLB

Let

$$Y_1, \dots, Y_n \stackrel{\text{i.i.d.}}{\sim} \text{Exp}(\beta), \quad f_Y(y | \beta) = \frac{1}{\beta} e^{-y/\beta}.$$

Find the CRLB for unbiased estimators of β .

The one-observation log-density is

$$\log f_Y(y | \beta) = \log(\beta^{-1}) - \frac{y}{\beta} = -\log \beta - \frac{y}{\beta}.$$

Differentiate once:

$$\frac{d}{d\beta} \log f_Y(y | \beta) = -\frac{1}{\beta} + \frac{y}{\beta^2}.$$

Differentiate twice:

$$\frac{d^2}{d\beta^2} \log f_Y(y | \beta) = \frac{1}{\beta^2} - \frac{2y}{\beta^3}.$$

Take expectation. Since $Y \sim \text{Exp}(\beta)$, $E[Y] = \beta$:

$$E \left[\frac{d^2}{d\beta^2} \log f_Y(Y | \beta) \right] = \frac{1}{\beta^2} - \frac{2E[Y]}{\beta^3} = \frac{1}{\beta^2} - \frac{2\beta}{\beta^3} = -\frac{1}{\beta^2}.$$

Therefore

$$I_n(\beta) = -n \left(-\frac{1}{\beta^2} \right) = \frac{n}{\beta^2}.$$

The CRLB is

$$\boxed{\text{Var}(\hat{\beta}) \geq \frac{1}{I_n(\beta)} = \frac{\beta^2}{n}.$$

5.2 Does $\hat{\beta}_{\text{MLE}} = \bar{Y}$ achieve the CRLB?

For the exponential distribution,

$$\text{Var}(Y_i) = \beta^2.$$

Thus

$$\text{Var}(\bar{Y}) = \text{Var} \left(\frac{1}{n} \sum_{i=1}^n Y_i \right) = \frac{1}{n^2} \sum_{i=1}^n \text{Var}(Y_i) = \frac{1}{n^2} n \beta^2 = \frac{\beta^2}{n}.$$

This equals the CRLB. Therefore:

$$\boxed{\bar{Y} \text{ is the MVUE for } \beta.}$$

Key point

General rules:

1. If an unbiased estimator achieves the CRLB, it is the MVUE.
2. The MVUE does **not** always achieve the CRLB. Sometimes no unbiased estimator reaches the bound.

6 Exponential Family Distributions

6.1 Definition

A family of distributions is an **exponential family** if its pdf or pmf can be written as

$$\boxed{f_Y(y | \theta) = a(\theta)b(y)e^{c(\theta)t(y)}}$$

where the support of $f_Y(y | \theta)$ does **not** depend on θ .

An equivalent form is

$$\boxed{f_Y(y | \theta) = b(y)e^{c(\theta)t(y)-d(\theta)}}.$$

In this second form,

$$a(\theta) = e^{-d(\theta)}.$$

Careful

The support condition matters. A density can look algebraically close to exponential-family form, but if its support depends on the parameter, it is not an exponential family under this definition.

6.2 Example: Exponential distribution

Let

$$Y \sim \text{Exp}(\beta), \quad f_Y(y) = \frac{1}{\beta} e^{-y/\beta}.$$

Use form

$$f_Y(y | \theta) = a(\theta)b(y)e^{c(\theta)t(y)}.$$

Set

$$a(\beta) = \frac{1}{\beta}, \quad b(y) = 1, \quad c(\beta) = -\frac{1}{\beta}, \quad t(y) = y.$$

Then

$$a(\beta)b(y)e^{c(\beta)t(y)} = \frac{1}{\beta} \cdot 1 \cdot e^{-y/\beta} = f_Y(y).$$

So the exponential distribution is an exponential family distribution.

6.3 Example: Binomial distribution

Let

$$Y \sim \text{Binomial}(n, p), \quad p_Y(y) = \binom{n}{y} p^y (1-p)^{n-y}.$$

Use form

$$p_Y(y | p) = b(y)e^{c(p)t(y)-d(p)}.$$

Rewrite:

$$\begin{aligned} p_Y(y) &= \binom{n}{y} p^y (1-p)^{n-y} \\ &= \binom{n}{y} e^{\log(p^y (1-p)^{n-y})} \\ &= \binom{n}{y} e^{y \log p + (n-y) \log(1-p)} \\ &= \binom{n}{y} e^{y[\log p - \log(1-p)] + n \log(1-p)} \\ &= \binom{n}{y} e^{y \log\left(\frac{p}{1-p}\right) + n \log(1-p)}. \end{aligned}$$

Therefore

$$b(y) = \binom{n}{y}, \quad c(p) = \log\left(\frac{p}{1-p}\right), \quad t(y) = y, \quad d(p) = -n \log(1-p).$$

So the binomial distribution is an exponential family distribution.

6.4 Example: Uniform distribution

Let

$$Y \sim \text{Unif}[0, \theta], \quad f_Y(y) = \frac{1}{\theta}, \quad 0 \leq y \leq \theta.$$

This is **not** an exponential family distribution under the definition above because the support depends on θ .

7 Sufficient Statistic for an Exponential Family

Let

$$Y_1, \dots, Y_n$$

be i.i.d. from an exponential family distribution

$$f_Y(y \mid \theta) = a(\theta)b(y)e^{c(\theta)t(y)}.$$

We want the sufficient statistic.

Recall the factorization criterion:

$$L(y \mid \theta) = g(u, \theta)h(y).$$

If all dependence on θ goes through u , then $U(Y) = u$ is sufficient.

Compute the likelihood:

$$\begin{aligned} L(y \mid \theta) &= \prod_{i=1}^n f_Y(y_i \mid \theta) \\ &= \prod_{i=1}^n a(\theta)b(y_i)e^{c(\theta)t(y_i)} \\ &= a(\theta)^n \left(\prod_{i=1}^n b(y_i) \right) e^{c(\theta) \sum_{i=1}^n t(y_i)}. \end{aligned}$$

This factors as

$$L(y \mid \theta) = \underbrace{a(\theta)^n e^{c(\theta) \sum_{i=1}^n t(y_i)}}_{g(u, \theta)} \underbrace{\prod_{i=1}^n b(y_i)}_{h(y)}.$$

Therefore the sufficient statistic is

$$U(Y) = \sum_{i=1}^n t(Y_i).$$

8 Exam-Use Procedure Summary

1. MLE of a function

If the question asks for the MLE of $g(\theta)$:

$$\widehat{g(\theta)}_{\text{MLE}} = g(\hat{\theta}_{\text{MLE}}).$$

Do not redo the likelihood unless you have not yet found $\hat{\theta}_{\text{MLE}}$.

2. MLE with parameter-dependent support

Write the likelihood with the support constraint. Then determine which boundary maximizes the likelihood.

Typical pattern:

$$L(\theta) \text{ decreases in } \theta \quad \Rightarrow \quad \hat{\theta} = \text{smallest value allowed by the data.}$$

3. Asymptotic distribution of the MLE

Use

$$\hat{\theta}_{\text{MLE}} \approx N\left(\theta, \frac{1}{nI(\theta)}\right).$$

Compute

$$I(\theta) = -\text{E}\left[\frac{d^2}{d\theta^2} \log f(Y \mid \theta)\right].$$

Then plug into the asymptotic Normal approximation.

4. CRLB

For unbiased estimators,

$$\text{Var}(\hat{\theta}) \geq \frac{1}{I_n(\theta)}.$$

If an unbiased estimator has variance equal to the CRLB, it is the MVUE.

5. Exponential family

Try to write the density or pmf as

$$f_Y(y \mid \theta) = a(\theta)b(y)e^{c(\theta)t(y)}$$

or

$$f_Y(y \mid \theta) = b(y)e^{c(\theta)t(y)-d(\theta)}.$$

Then check support. If support depends on θ , it fails this definition.

For an i.i.d. exponential family sample,

$$U(Y) = \sum_{i=1}^n t(Y_i)$$

is sufficient.